ORF 245: Fundamentals of Statistics – Spring 2025

Precept 1

Last updated: January 31, 2025

1. (**Ice breaking + Introductions**)
   - Introduce yourself.
   - Set communication ground rules: e.g., can stick around for up to 10 minutes after precept. Additional help from office hours.

2. (**Set Theory**)

   (a) Formal and informal ways of describing sets
   - $\{2, 4, 6, 8, 10, \ldots\}$: The set of all positive even integers.
   - $\{\ldots, -3, -1, 1, 3, \ldots\}$: The set of all odd integers.
   - $\{n | n = 2m \text{ for some } m \in \mathbb{N}\}$: The set of all positive even integers
   - $\{x | x = 2n \text{ and } x = 3m \text{ for some } n, m \in \mathbb{N}^+\}$: The set of all positive multiples of 6.
   - $\{b | b \in \mathbb{Z}, b = b + 1\}$: empty set

   (b) Set operations: subset, union $\cup$, intersection $\cap$, complement c. sample space $S = \{x, y, z\}$, $A = \{x, y\}$, $B = \{x, z\}$
   - A, B are all subsets of S.
   - $A \cup B = B \cup A = \{x, y, z\}$
   - $A \cap B = \{x\}$
   - $A^c = \{z\}$

   (c) Venn diagram: show $A, B, A \cap B, A \cup B$, A^c, $B \cap A^c$.

3. (**Probability Basics Review**)

   Basic Concepts with example
   - *Experiment*, *Outcome* (e.g. throw a 6-sided die twice)
   - *Sample Space* S: the set of all outcomes of an experiment.
     (e.g. $\{(1, 1), ..., (1, 6); (2, 1), ..., (2, 6); (6, 1), ..., (6, 6)\}$)
   - *Event* A: a collection of outcomes (a subset of S).
   - Some example: $A = \{\text{get a sum of at most } 10\}$, $B = \{\text{throw a 1 on the first die OR 5 on the second die}\}$.

   Probability
   - Three Axioms of Probability

     (a) (Positivity) For any event A, $\mathbb{P}(A) \geq 0$.

     (b) (Normalization) $\mathbb{P}(S) = 1$.

     (c) If $A_1, A_2, ...$ is an countably infinite collection of disjoint events, then

     $$\mathbb{P}\left(\bigcup_{k=1}^{\infty} A_k\right) = \sum_{k=1}^{\infty} \mathbb{P}(A_k)$$

- Intuition: assigned non-negative weights on all events of interest.
- The above example, two different weights: fair die $\mathbb{P}_1$, unfair die $\mathbb{P}_2$ (1/2 probability on 1, 1/10 probability on others).

  Note: $\mathbb{P}_1$ and $\mathbb{P}_2$ are two different probability measures.

Properties of Probability

(a) $\mathbb{P}(\emptyset) = 0$.

(b) $\mathbb{P}(A) = 1 - \mathbb{P}(A^c)$.

(c) (Inclusion-Exclusion Principle) $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

(d) (Law of Total Probability) $\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c)$.

Calculation of Probability

- Combine definition with properties.
- (Use (b)): $\mathbb{P}_1(A) = 11/12$, $\mathbb{P}_2(A) = 97/100$.
- (Use (c)) $\mathbb{P}_1(B) = 11/36$, $\mathbb{P}_2(B) = 11/20$.
  - **Solution**:
    * Use IE Principle, define $C = \{\text{throw a 1 on first die}\} = \{(1, x) : x \in [6]\}$, $D = \{(x, 5) : x \in [6]\}$, $C \cap D = \{(1, 5)\}$. Need to calculate $\mathbb{P}_k(C \cap D)$, $\mathbb{P}_k(C)$ and $\mathbb{P}_k(D)$
    * $\mathbb{P}_1(C) = 1/6$, $\mathbb{P}_2(C) = 1/2$; $\mathbb{P}_1(D) = 1/6$, $\mathbb{P}_2(D) = 1/10$; $\mathbb{P}_1(C \cap D) = 1/36$, $\mathbb{P}_2(C \cap D) = 1/20$.
    * $\mathbb{P}_1(C \cup D) = 11/36$, $\mathbb{P}_2(C \cup D) = 11/20$.